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Abbott

BI & Data Engineering

Case Study

Scalable GCP-native Data Pipelines + Power BI / Looker Self-Serve Analytics for Global Healthcare & Diagnostics Operations

HEALTHCARE

GCP DATA ENGINEERING

POWER BI | LOOKER

GLOBAL OPERATIONS

*Unifying Abbott's Diagnostics, Pharma & Commercial Data on GCP with a Power BI / Looker Self-Serve Analytics Platform***4x**Faster Pipeline
Throughput**25+**GCP Services
Integrated**65%**Infra Cost
Reduction**1500+**Concurrent Dashboard
Users**Problem Statement**

- ▶ Siloed SAP, Veeva & LIMS data on legacy systems
- ▶ No real-time diagnostics analytics on cloud
- ▶ Static reports with no self-serve capability
- ▶ Fragmented BI tools per department

Solution Overview

- ▶ GCP-native lakehouse (BigQuery + Dataflow)
- ▶ Cloud Composer DAG orchestration
- ▶ Power BI + Looker dashboards
- ▶ React.js + Node.js analytics portal

Business Value

- ▶ Real-time batch release & QC dashboards
- ▶ 4x pipeline throughput improvement
- ▶ Self-serve analytics for 1500+ global users
- ▶ 65% infra cost reduction vs on-prem

O u t c o m e

Data Engineering & Pipeline Speed

GCP Dataflow + Cloud Composer pipelines replaced legacy Informatica jobs — achieving 4x pipeline throughput and reducing batch processing windows from 10 hours to under 2 hours.

Cost Optimisation

Decommissioned 6 on-premise Hadoop clusters by migrating to BigQuery + GCS — achieving 65% reduction in infrastructure costs with auto-scaling for fiscal quarter-end peaks.

Frontend & User Experience

Built a Power BI + React.js analytics portal with role-based access, 20 custom dashboards, and D3.js visualisations — enabling 1500+ concurrent users with sub-1.8s page load times.

Regulatory & Compliance

Automated FDA 21 CFR Part 11, GxP, and pharmacovigilance audit trail generation through GCP Dataform + BigQuery — cutting compliance report turnaround from T+7 to same-day.

Problem Statement

Abbott operates across 50+ markets with a complex tech stack spanning SAP S/4HANA, Veeva Vault, LIMS (LabWare), MES (Syncade), Salesforce CRM, and third-party market data feeds. Analytics relied on on-premise Hadoop clusters and static Excel/SSRS reports — with no modern self-serve BI, no real-time diagnostics visibility, and significant compliance risk.

01

Legacy On-Prem Hadoop Infrastructure

6 on-premise Hadoop clusters running Informatica ETL jobs — causing 10-hour batch windows, frequent job failures, and no elasticity for peak processing loads.

02

Fragmented Data Sources & No Cloud Ingestion

SAP S/4HANA, Veeva Vault, LIMS, MES, and Salesforce CRM data were siloed with no unified GCP ingestion layer — causing inconsistent KPIs across commercial and diagnostics teams.

03

No Modern Self-Serve Analytics Portal

Analytics were delivered via static SSRS/Excel reports emailed to teams — no interactive dashboards, no self-serve filtering, and no mobile-responsive access for field reps.

04

Compliance & GxP Audit Trail Gaps

FDA 21 CFR Part 11, GxP, and pharmacovigilance reports required manual collation from 7+ systems — creating T+7 delays and high risk of regulatory submission errors.

Solution Overview

Source Systems

SAP S/4HANA | Veeva Vault | LIMS (LabWare) | MES (Syncade) | Salesforce CRM | Market Data Feeds

CLOUD PUB/SUB + CLOUD DATAFLOW

Stream & Batch Ingestion | CDC Pipelines | Schema Validation (Avro/Protobuf) | Dead-Letter Queues

CLOUD COMPOSER (Airflow DAGs) + DATAPROC SPARK

ETL Orchestration | DQ Checks | SLA Alerting | Incremental & Full-Refresh Loads

GCS Lakehouse — Bronze → Silver → Gold

Partitioned + Tagged | Dataplex Governance | Auto-Schema Evolution | Policy Tags

BIGQUERY DWH + DATAFORM

SQL Transforms + Lineage | Star Schema | Materialised Views | Row-Level Security

POWER BI + LOOKER + REACT/NODE.JS PORTAL

Commercial | Diagnostics | Regulatory | QC | Doctor 360 | Supply Chain

12+ TB Data Migrated

50+ Markets

1500+ Concurrent Users

25 GCP Services

200+ Pipelines

G C P D a t a E n g i n e e r i n g D o m a i n

Ingestion & Streaming

- ◆ Cloud Pub/Sub for SAP & Salesforce change events
- ◆ Dataflow Apache Beam streaming jobs
- ◆ Dead-letter queue + alerting via Cloud Monitoring
- ◆ Schema registry with Protobuf/Avro validation

Batch ETL & Transformation

- ◆ Cloud Composer Airflow DAGs (200+ pipelines)
- ◆ Dataproc Spark for complex healthcare data joins
- ◆ Dataform SQL models with unit test framework
- ◆ Incremental & full-refresh load strategies

GCS Lakehouse Architecture

- ◆ Bronze/Silver/Gold zone partitioning on GCS
- ◆ Delta-equivalent using BigQuery time-travel
- ◆ Dataplex for zone governance & data quality
- ◆ Auto-schema evolution with policy tags

BigQuery Analytics DWH

- ◆ Star schema for Commercial & Diagnostics domains
- ◆ Materialised views for sub-second dashboard queries
- ◆ Slot reservations for guaranteed QoS at peak load
- ◆ Row-level security by market & business unit

ML & Vertex AI

- ◆ Patient demand forecasting on Vertex AI Workbench
- ◆ Feature Store for reusable healthcare ML features
- ◆ AutoML for batch QC anomaly detection
- ◆ Model monitoring & drift detection pipelines

Governance & Observability

- ◆ Dataplex automated data classification & tagging
- ◆ Data Catalog with 800+ business attribute entries
- ◆ Cloud Monitoring dashboards + PagerDuty alerts
- ◆ End-to-end lineage from SAP to BigQuery table

BI & Analytics Domain

BI Architecture & Engineering Pillars

Power BI Semantic Layer

DirectQuery mode on BigQuery for live data; composite models with aggregations; Row-level security via Azure AD groups; incremental refresh policies.

Looker on GCP (LookML)

Governed LookML data models with PDTs; Single Looker Explore powering multiple dashboards; custom fiscal calendar dimension for healthcare periods.

React.js + Node.js Portal

React 18 + TypeScript SPA; lazy-loaded modules reducing bundle size by 55%; Node/Express API with JWT auth, RBAC, rate limiting & BigQuery connection pooling.

Security & Deployment

Cloud Run containerised deployment with Cloud Armor WAF; OAuth 2.0 + Google Identity SSO; Cloud CDN for static assets; 99.97% SLA uptime.

20 Dashboard Modules Delivered

Commercial Sales & Market Share

Commercial

Doctor 360 & Rx Trends

CRM

Territory & Rep Performance

Field Force

Diagnostics Batch Release Tracker

Mfg

QC / QA Deviation & CAPA Monitor

Quality

Supply Chain & Inventory Health

SCM

Demand Forecasting & S&OP

Planning

Regulatory Submission Tracker

Compliance

Financial P&L Dashboard

Finance

ESG & Sustainability Metrics

ESG

Ganit's Deliverables and the Value Generated for Abbott

D e l i v e r a b l e s & V a l u e G e n e r a t e d

GCP Platform Setup & Migration

Provisioned GCP landing zone with VPC, IAM, Dataplex zones, and migrated 12 TB of historical data from 6 on-premise Hadoop clusters to GCS Bronze layer.

Data Pipeline Engineering

Built 200+ Cloud Composer DAGs and Dataflow jobs ingesting SAP S/4HANA, Veeva Vault, LIMS, MES, Salesforce CRM — with DQ rules, retry logic, and SLA monitoring.

BigQuery Data Warehouse

Designed star schema models with Dataform SQL transforms, materialised views, row-level security, and slot reservations for guaranteed performance at quarter-end peaks.

Power BI + Looker + React/Node.js Portal

Delivered a 20-module analytics platform with Power BI semantic layer, LookML models, JWT auth RBAC, D3.js/Recharts, REST API (Node/Express), Cloud Run deployment.

Governance & Observability

Implemented Dataplex data catalog with 800+ attributes, Cloud Monitoring dashboards, PagerDuty alerting, end-to-end lineage, and FDA 21 CFR Part 11 + GxP audit automation.

4x Faster

Pipeline Throughput

Replaced 10-hour Informatica batch jobs with GCP Dataflow streaming — delivering data to BigQuery in under 2 hours.

65% Lower

Infrastructure Cost

Decommissioned 6 Hadoop clusters — migrating to serverless BigQuery + GCS with auto-scaling for cost-efficient peak processing.

<1.8s Load

Dashboard Response

Materialised views + partitioned BigQuery tables + Cloud CDN delivered sub-1.8-second P95 load times across all 20 modules.

1500+

Concurrent Portal Users

Cloud Run autoscaling + Node.js connection pooling supports 1500+ simultaneous users with zero performance degradation.

Same-Day

Compliance Reporting

FDA 21 CFR Part 11, GxP, and pharmacovigilance reports automated end-to-end — replacing T+7 manual collation processes.

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Thank You

Ganit provides cutting edge solutions at the intersection of hypothesis-based analytics, ML-driven AI, and new-data insights from smart devices

We have delivered tangible benefits to leading brands in North America, Europe, MENA and APAC

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